

Useful Information

- The solutions to the quadratic equation $ax^2 + bx + c = 0$ are $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
- In the function $f(t) = Ae^{-t/\tau}$ the term τ is sometimes referred to as the time constant.
- The following Laplace transform pairs may be of use. Here $u(t)$ is the unit step function

$$\begin{aligned}
 u(t) &\longleftrightarrow \frac{1}{s} \\
 e^{-at}u(t) &\longleftrightarrow \frac{1}{s+a} \\
 \sin(\omega t)u(t) &\longleftrightarrow \frac{\omega}{s^2 + \omega^2} \\
 \cos(\omega t)u(t) &\longleftrightarrow \frac{s}{s^2 + \omega^2} \\
 e^{-at} \cos(\omega t)u(t) &\longleftrightarrow \frac{s+a}{(s+a)^2 + \omega^2} \\
 e^{-at} \sin(\omega t)u(t) &\longleftrightarrow \frac{\omega}{(s+a)^2 + \omega^2} \\
 t^n u(t) &\longleftrightarrow \frac{n!}{s^{n+1}} \\
 \frac{df(t)}{dt} &\longleftrightarrow sF(s) - f(0^-) \\
 \frac{d^2f(t)}{dt^2} &\longleftrightarrow s^2F(s) - sf(0^-) - f'(0^-) \\
 \int_{0^-}^t f(\lambda) d\lambda &\longleftrightarrow \frac{F(s)}{s}
 \end{aligned}$$

- A generic form for a rational Laplace Transform is $G(s) = K_G \frac{\prod_{i=1}^M (s+z_i)}{\prod_{j=1}^n (s+p_j)}$, where the set of finite zeros is $\{-z_i\}_{i=1}^M$ and the set of poles is $\{-p_j\}_{j=1}^n$.
- The final value theorem states that for a Laplace Transform $Y(s)$ with all its poles in the left half plane, except for at most one pole at the origin, $\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$.
- For a standard second-order system with a transfer function of the form $T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$, when $\zeta < 1$, the poles are at positions $-\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$. The step response of such a system has the following properties:

- $y_{\text{step}}(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n\sqrt{1-\zeta^2}t + \theta)$, where $\theta = \cos^{-1}(\zeta)$.
- 2% settling time, $T_s \approx \frac{4}{\zeta\omega_n}$.
- Percentage (maximum) overshoot, $P.O. = 100e^{-\pi\zeta/\sqrt{1-\zeta^2}}$. Hence, $\zeta = \frac{-\ln(P.O./100)}{\sqrt{\pi^2 + \ln^2(P.O./100)}}$.
- Time of peak, $T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$.
- For $0.3 \leq \zeta \leq 0.8$, the 10%-90% rise time is $T_r \approx \frac{2.16\zeta + 0.6}{\omega_n} \approx \frac{2}{\omega_n}$.

- Steady state error constants:
 - Position error constant: $K_{\text{posn}} = \lim_{s \rightarrow 0} G_c(s)G(s)$
 - Velocity error constant: $K_v = \lim_{s \rightarrow 0} sG_c(s)G(s)$
 - Acceleration error constant: $K_a = \lim_{s \rightarrow 0} s^2G_c(s)G(s)$